

YVETTE D. HASTINGS

DATA SCIENTIST

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PROFESSIONAL SUMMARY

Data scientist with experience transforming complex datasets into actionable business insights through statistical analysis, machine learning, predictive modeling, and interactive visualization. Experienced designing end-to-end analytical workflows, communicating findings to technical and non-technical stakeholders, and developing dashboards that support data-driven decision making. Recognized for data integrity, analytical rigor, adaptability, and attention to detail.

TECHNICAL SKILLS

- **Data Science Methods:** Exploratory Data Analysis, NLP, Time Series Analysis, Machine Learning, Statistical Modeling, Experimental Design, Data Visualization, Database Management
- **Data Analysis Tools:** R, Python, Java, SQL, Excel
- **Machine Learning & Analytics:** scikit-learn, pandas, NumPy, BeautifulSoup, rvest, dplyr, tidyverse, matplotlib, Plotly, ggplot
- **Reporting Tools:** Overleaf, MS Office Suite, Adobe Creative Cloud
- **Version Control:** Git, GitHub, GitLab

DATA SCIENCE PROJECTS

Gallatin Housing Predictions

- Engineered predictive features through data cleaning, one-hot encoding, outlier detection, missing value handling, and feature standardization prior to model development.
- Applied exploratory data analysis, statistical hypothesis testing, and 5-fold cross-validation to evaluate predictive model performance on a Zillow dataset containing over 17,000 Gallatin County residential property records.
- Developed and evaluated regression, clustering, and classification models using Python and scikit-learn to predict factors influencing home prices in Gallatin County, MT.
- Achieved $R^2 = 0.608$ and $RMSE = 0.79$ using Lasso and Ordinary Least Squares regression models.

Cybersecurity Incident Modeling

- Developed regression and classification models using Python and scikit-learn to predict cybersecurity financial loss and incident resolution outcomes.
- Applied feature engineering, PCA, hypothesis testing, and cross-validation workflows on synthetic cybersecurity datasets.
- Achieved $R^2 = 0.7449$ using Gradient Boosting and 78% classification accuracy using Logistic Regression.

Public Park Accessibility Analysis

- Applied data cleaning techniques to geospatial big data for analysis and visualization in R and ArcGIS Pro.
- Queried and integrated U.S. Census database records using SQL to evaluate socioeconomic accessibility to public parks.

WORK EXPERIENCE

Graduate Research Assistant

Montana State University | Bozeman, MT | January 2023 – May 2026

- Applied statistical analysis and machine learning using R and Python to evaluate research hypotheses and identify significant predictors across environmental and software engineering datasets.

- Designed and implemented a web-based visualization platform using React to improve user interaction with software quality assessment data.
- Presented analytical findings through technical reports, interactive dashboards, and peer-reviewed publications.
- Mentored 4+ students in cybersecurity concepts, with practical applications and experiences in digital forensics tools.

Graduate Research & Teaching Assistant

The University of Utah | Salt Lake City, UT | August 2020 - December 2022

- Designed and developed an environmental monitoring dashboard using R Shiny to improve citizen science and research collaboration.
- Executed in-depth studies on surface water and urban stream system interactions, employing R for data analysis and interpretation.
- Applied data science principles to perform statistical analysis and data interpretation of soil enzyme interactions.

Inorganic Chemistry Department Assistant Supervisor

American West Analytical Laboratory | Salt Lake City, UT | July 2016 – July 2020

- Trained 30+ new and current employees on EPA and laboratory methods to enhance performance and efficiency.
- Research and apply EPA methods to create and update laboratory SOPs.
- Review department analytical data for final client reports ensuring compliance with EPA and other accrediting agency requirements.
- Streamlined workflow processes to increase overall laboratory efficiency.

EDUCATION

Master of Science in Computer Science

Montana State University | Bozeman, MT | January 2023 – August 2026

- **Selected Publications:**
 - Y.D. Hastings et al. (2026), “Barriers to Use: Perspectives on Environmental Research Software”, in *Proceedings of the 21st International Conference on Software Technologies (ICSOFT 2026)*, SciTePress, pp. 87-38.
 - Distributed and analyzed surveys to determine factors influencing user perceived software quality.
 - Y.D. Hastings et al. (2026), “Mining User Forums to Evaluate Quality-in-Use of Environmental Software”, in *SoutheastCon 2026*, IEEE, pp. 1-8.
 - Collected, cleaned, and analyzed large text datasets using web scraping, natural language processing, and statistical analysis on publicly available online user forms.
 - Performed NLP and statistical analysis to determine factors influencing user perceived software quality.
- **Relevant Coursework:** Empirical Software Engineering, Advanced Software Engineering, Advanced Data Mining, Data Storytelling

Master of Science in Geography

The University of Utah | Salt Lake City, UT | August 2020 – August 2022

- **Thesis:** “Green Infrastructure Microbial Community Response to Simulated Pulse Precipitation Events in the Semi-Arid Western U.S.”
- **NSF Grant:** “Can green infrastructure maximize ecosystem processes related to nitrogen?” (NSF DEB 2006308)
- **Selected Publications:**
 - Y.D. Hastings et al. (2023), “Green infrastructure microbial community response to simulated pulse precipitation events in the semi-arid western united states,” *Water*, vol. 16, no. 13, p. 1931.
 - Performed advanced statistical methods to determine leading environmental drivers influencing enzyme activity rates and nitrogen regulation in soil.
- **Relevant Coursework:** Advanced Geographic Data Analysis, Geospatial Big Data, Data Visualization, Spatial Modeling and Geocomputation

Bachelor of Science in Environmental & Sustainability Studies

The University of Utah | Salt Lake City, UT | August 2016 – July 2020